

DATA PORTFOLIO: LESSONS LEARNED

Project: Bash Automation & Log Analysis Workflow

1. Environment & Architecture

- **Native Linux Performance:** Projects stored in `/mnt/c/` suffer performance hits. Native paths (`~/portfolio`) are significantly faster and resolve permission-based visual glitches (the "bright green" highlight).
- **WSL Defaults:** Docker Desktop often steals the default WSL slot. Use `wsl --set-default Ubuntu` to restore your primary dev environment.
- **Server Sockets:** The `ENOENT` socket error occurs when the editor's background server loses sync; clearing `~/.cursor-server` forces a healthy reset.

2. Scripting Logic & Syntax

Variable Best Practices

- **Expansion:** Do NOT quote tildes in assignments. `LOG_DIR=~/path` works; `LOG_DIR=~/"path"` breaks because Bash treats `~` as a literal character.
- **Sanitization:** Always double-quote variables in usage (`"$LOG_DIR"`) to protect against spaces in paths.
- **Typos:** A single character error (`$LG_DIR`) evaluates to null, causing the script to search root instead of the target folder.

3. Command & Utility Mastery

- **The find Command:** Wildcards like `*.log` must be quoted (`"*.log"`) to prevent the shell from expanding them before the command executes.
- **Relative Navigation:** `..` (parent) and `.` (current) are essential for moving data without typing absolute paths (e.g., `mv ../* .`).
- **Modes:** `chmod +x` is the "key" that turns a text file into an executable program.

4. Advanced Automation Patterns

```
# Logic: Nested loop for multi-log analysis
for log in "${log_files[@]}; do
  for pattern in "${patterns[@]}; do
    # Count and record findings
    count=$(grep -c "$pattern" "$log")
    if [ "$count" -gt 10 ]; then
      echo "Action Required: High $pattern count!"
    fi
  done
done
```

- **Redirection:** `>` (overwrite) vs `>>` (append) is the difference between a clean report and a continuous log history.
- **Intelligent Alerting:** Scripts should do more than search; they should evaluate. Threshold-based `if` statements turn data into actionable intelligence.